

Rotation Matrices Part 2 and HT Matrices

EGR 445/545

GVSU

Plan for tonight

- review:
 - rotation signs and conventions
 - unit vectors as columns
 - dot products
- multiple rotations
- intro to HT matrices
- project 1 kick-off
 - two-link inverse kinematics
- LA 3 and 4

Announcements

- Quiz 1 next Wednesday, 5/27
 - what is a rotation matrix?
 - where do rotation matrices come from?
 - how do I find an expression for one based on a sketch?
 - what does it mean to say the unit vectors are the columns?
- Chapter 2 Learning Outcomes posted

Review

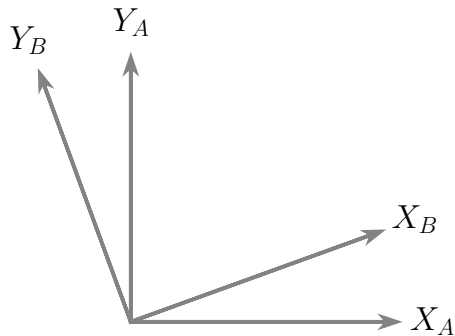
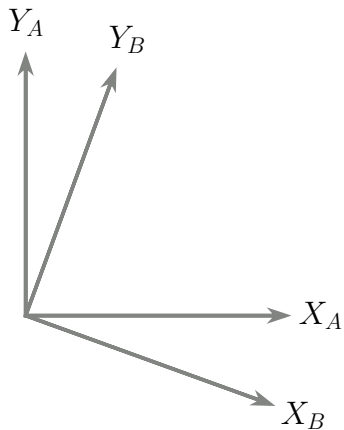
Conceptual Review

- why do we need to learn rotation matrices?
- why do we also need HT matrices?

Steps and Sign Conventions

- what are the steps and sign conventions in finding ${}^A_B\mathbf{R}$?

Which of these is a positive rotation?



Dot Products

- what is the relationship between unit vectors and rotation matrices?

Finding ${}^A_B\mathbf{R} = \mathbf{R}_y(\theta)$

- what sketch do we need to draw?
 - what does a positive $\mathbf{R}_y$ rotation look like?
- which unit vectors do we need to draw?
 - the unit vectors map from which frame to which frame?

Multiple Rotations: the Notation is Your Friend

- step 1: first, we rotate frame 1 from frame 0 by θ_1
- step 2: then, we rotate frame 2 from frame 1 by θ_2
- if we know 2P , how do we find 0P ?

Finding $\mathbf{R}$ with multiple rotations

- step 1: first, we rotate frame 1 from frame 0 by θ_1
- step 2: then, we rotate frame 2 from frame 1 by θ_2
- how do we find ${}^0_2\mathbf{R}$?

$${}^0_1\mathbf{R} \cdot {}^1_2\mathbf{R} = ?$$

$${}^1_2\mathbf{R} \cdot {}^0_1\mathbf{R} = ?$$

Multiple Rotations and Order of Rotations

Let the notation guide you

Key question:

- If we were going to rotate first from A to B and then from B to C, how would we find ${}^A_C\mathbf{R}$?
 - can our notation conventions help guide us?
 - what does the math behind ${}^A_C\mathbf{R}$ really mean?

Remaining Chapter 2 Topics

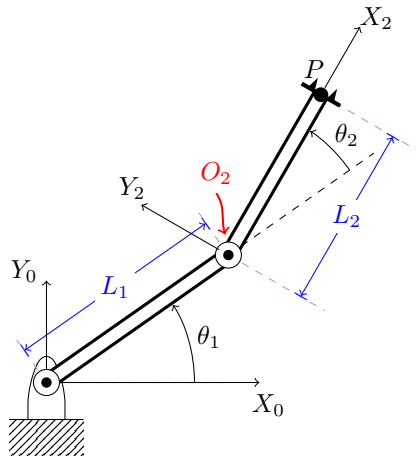
- how do we handle rotation and translation together?
- what is an HT matrix and how do they work?
- how do we find the inverse of an HT matrix?
- do rotation matrices really have 9 unknowns?
- constraints and Euler angles
- what are free vectors and how to we deal with them?
- computational considerations

Intro to HT Matrices

Intro to HT Matrices

- How do we find ${}^1P_{tip}$?
- what is wrong with this?:

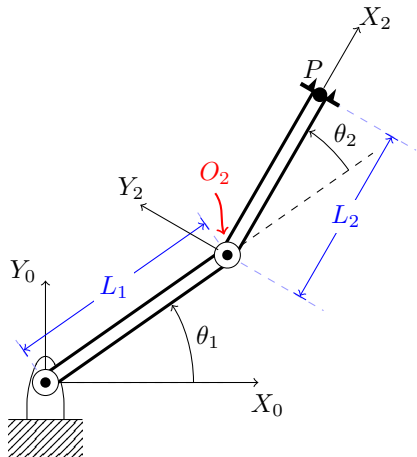
$${}^1P_{tip} = {}^1_2\mathbf{R} \cdot {}^2P_{tip}$$



The point of an HT matrix

- how do we put this in one matrix operation?

$${}^1P_{tip} = {}^1P_{O2} + {}^1_2\mathbf{R} \cdot {}^2P_{tip}$$



The Form of an HT Matrix

- how do we add a translation and do a rotation in one step?

$${}^0_1\mathbf{T} = ?$$

Project 1 Kick-Off

https://drive.google.com/file/d/1Q2ajgSWEvCEuOxaHtxc_tfSVt97T0Ubf/view?usp=sharing

- projects folder under student gdrive folder

Two-Link Inverse Kinematics

`https://drive.google.com/drive/folders/
1LH1sVy-XgVXGyPA46GKrik_rb-pbArlN?usp=sharing`

- under class folders

Active Learning

- learning activities 4 and 5